Advanced Level Statistics

Statistical Limit Theorems and Probability Distributions

1. Law of Large Numbers (LLN)

Definition

> *As the number of trials increases, the **sample mean approaches the population mean**.*

If X_1, X_2, X_3, ..., X_n are independent random variables, the sample mean is:

$$\bar{X} = (1/n) \times \Sigma X_i$$

As $n \to \infty$:

$$\bar{X} \to \mu$$

```
μ = population mean
```

Interpretation

When an experiment is repeated many times:

- the **average result stabilizes**
- it becomes **close to the expected value**

Example

A fair coin is tossed many times:

```
Probability of head = 0.5
```

Number of Tosses	Proportion of Heads
10	0.6
100	0.52
1000	0.501

As the number of tosses increases:

```
Proportion of heads → 0.5
```

Concept Diagram

```
Number of Trials → Large
        |
        v
    Sample Mean
```

```
        |
        v
   Population Mean
```

Sample Problem 1

A coin is tossed **2000 times** and heads appear **1020 times**. Find the experimental probability of heads.

Solution

$$P(H) = \text{Number of heads} / \text{Total tosses}$$

$$P(H) = 1020 / 2000 = 0.51$$

This value is **close to the theoretical probability 0.5**, illustrating the **Law of Large Numbers**.

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2. Distributions Derived from Normal Distribution

Some important statistical distributions are derived from the **normal distribution**. These are widely used in **statistical hypothesis testing**:

- Chi-square distribution
- t distribution
- F distribution

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3. Chi-Square Distribution (χ^2)

Definition

If $Z \sim N(0,1)$, then:

$$U = Z^2$$

follows a **Chi-square distribution with 1 degree of freedom**. More generally:

$$\chi^2 = Z_1^2 + Z_2^2 + \dots + Z_k^2$$

follows a **Chi-square distribution with k degrees of freedom**.

Properties

- Always **positive**
- Skewed to the **right**
- Depends on **degrees of freedom**

Applications

Chi-square distribution is used in:

- Goodness-of-fit tests
- Independence tests
- Hypothesis testing

Example

Suppose observed frequencies differ from expected frequencies in a survey. The **Chi-square statistic** is:

$$\chi^2 = \Sigma\,(O - E)^2 / E$$

```
O = observed frequency
E = expected frequency
```

Sample Problem 2

Observed and expected frequencies — compute χ^2.

```
O = {20, 30, 50}
E = {25, 25, 50}
```

Solution

$$\chi^2 = (20–25)^2/25 + (30–25)^2/25 + (50–50)^2/50$$

$$\chi^2 = 25/25 + 25/25 + 0$$

$$\chi^2 = 1 + 1 = 2$$

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4. t Distribution

Definition

If:

```
Z ~ N(0,1)
U ~ χ² with n degrees of freedom
```

then:

$$T = Z / \sqrt{(U/n)}$$

follows the **t distribution with n degrees of freedom**.

When t Distribution is Used

The **t distribution** is used when:

- the population variance is **unknown**
- the **sample size is small**

Properties

- Symmetrical like normal distribution
- Has heavier tails
- Approaches normal distribution as sample size increases

Sample Problem 3

Given:

```
x■ = 52  (sample mean)
μ = 50  (population mean)
s = 8   (standard deviation)
n = 16  (sample size)
```

Compute t.

Solution

$$t = (X\blacksquare - \mu) / (s / \sqrt{n})$$

$$t = (52 - 50) / (8 / \sqrt{16})$$

$$t = 2 / (8/4) = 2 / 2 = 1$$

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5. F Distribution

Definition

If $U \sim \chi^2(m)$ and $V \sim \chi^2(n)$ are independent variables, then:

$$F = (U/m) / (V/n)$$

follows the **F distribution with m and n degrees of freedom**.

Properties

- Always positive
- Used to compare **two variances**

Applications

F distribution is used in:

- ANOVA (Analysis of Variance)
- Comparing two population variances
- Regression analysis

Sample Problem 4

Given two sample variances, compute F-statistic:

```
s₁² = 16
s₂² = 9
```

Solution

$$F = s_1^2 / s_2^2$$

$$F = 16 / 9 = 1.78$$

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6. Sampling Distribution

A **sampling distribution** describes the probability distribution of a statistic obtained from repeated samples. Common sampling statistics:

- Sample mean
- Sample variance

Sample Mean

$$\bar{X} = (1/n) \Sigma X_i$$

Properties:

$$E(\bar{X}) = \mu$$

$$Var(\bar{X}) = \sigma^2 / n$$

Sample Variance

$$S^2 = (1/(n-1)) \Sigma (X_i - \bar{X})^2$$

It measures how much the sample values vary around the sample mean.

Standard Error of Mean

$$SE = \sigma / \sqrt{n}$$

Example

```
Sample size n = 36
Population mean μ = 80
Population standard deviation σ = 18
```

$$SE = 18 / \sqrt{36} = 18 / 6 = 3$$

Sample Problem 5

A population has $\mu = 50$, $\sigma = 10$. A sample of size $n = 25$. Find the standard error.

Solution

$$SE = \sigma / \sqrt{n}$$

$$SE = 10 / \sqrt{25} = 10 / 5 = 2$$

Advanced Level Summary

Topics Covered

- ✓ Law of Large Numbers
- ✓ Chi-square distribution
- ✓ t distribution
- ✓ F distribution
- ✓ Sampling distributions
- ✓ Sample mean and variance
- ✓ Standard error